

ARUN KUMAR REDDY N

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SUMMARY

~7 YoE optimizing **Computer Vision** algorithms - from implementing high-performance parallelized software on **DSP** and **GPU** to developing **Firmware** for **Custom Silicon Accelerators**

EXPERIENCE

Qualcomm

Optimization Engineer

Feb 2025 - Present

- **Parallelized XR Algorithms** on **Qualcomm Hexagon DSP** using **HVX SIMD** intrinsics and **Halide** framework to accelerate performance
- Abstracted a **C++26 std::simd** style interface to low-level SIMD operations, driving team-wide **SIMD Adoption** and hence significantly improving **Hardware Utilization**
- Debugged **Hardware Bugs** to maintain SIMD acceleration, preventing inefficient scalar fallbacks

Firmware Engineer

Sept 2022 - Jan 2025

- Developed **Firmware** for CV **Hardware Accelerator** across the full product lifecycle from FPGA verification to commercialization in Mobile, Auto and IoT
- Implemented major firmware architecture changes, including **Dynamic HW Scheduling** and **DMA Utilities** to improve hardware utilization
- Led **Competitive Analysis** by developing and bringing up **Multi-Threaded Firmware** on **Hexagon DSP** to evaluate it as a controller

Valeo

Optimization Engineer

July 2019 - Sept 2022

- Ported **ADAS CV** algorithms to heterogeneous architectures, including **Renesas DSP**, **Qualcomm Adreno GPU (OpenCL)** and **ARM NEON**
- **CMake** based cross-compilation to evaluate CV algorithm prototypes on **Nvidia** and Qualcomm SoCs
- Automated a processor-in-loop framework to expedite issue reproduction and debugging on target SoC

EDUCATION

Ramaiah Institute Of Technology

B.E. (Electrical & Electronics Engineering)

Aug 2015 - June 2019

- FAER McAfee National Scholarship Award Program 2018-19
- NCC ('C' & 'B' Certificate), NSS (2 years)

SKILLS

C, C++ | SIMD (NEON, HVX) | DSP (Renesas, Hexagon) | GPGPU (OpenCL, CUDA) | Firmware | Multi-Threading |